

Exercise 1-7

Q

Question: 25-26

Find all values of unknown constant for which A is symmetric

25-

$$A = \begin{bmatrix} 4 & -3 \\ a+5 & -1 \end{bmatrix}$$

As A is Symmetric

$$A = A^T$$

$$\begin{bmatrix} 4 & -3 \\ a+5 & -1 \end{bmatrix} = \begin{bmatrix} 4 & a+5 \\ -3 & -1 \end{bmatrix}$$

compare

$$a+5 = -3$$

$$a = -3-5$$

$$\boxed{a = -8}$$

26-

$$A = \begin{bmatrix} 2 & a-2b+2c & 2a+b+c \\ 3 & 5 & a+c \\ 0 & -2 & 7 \end{bmatrix}$$

AS A is Symmetric

$$A = A^T$$

$$\begin{bmatrix} 2 & a-2b+2c & 2a+b+c \\ 3 & 5 & a+c \\ 0 & -2 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 0 \\ a-2b+2c & 5 & -2 \\ 2a+b+c & a+c & 7 \end{bmatrix}$$

compare

$$a-2b+2c=3 \rightarrow (1) \quad a+c=-2 \rightarrow (3)$$

$$2a+b+c=0 \rightarrow (2)$$

Use eq (3)

$$a+c=-2$$

$$c = -2-a \rightarrow (4)$$

put $c = -2-a$ in eq (2)

$$2a+b-2-a=0$$

$$a+b-2=0$$

$$a+b=2 \rightarrow (5)$$

put $c = -2-a$ in eq (1)

$$a-2b+2(-2-a)=3$$

$$a-2b-4-2a=3$$

$$-a-2b=7 \rightarrow (6)$$

Add eq (5) and (6)

$$+ a+b=2$$

$$+ -a-2b=7$$

$$-b=9$$

$$\boxed{b = -9}$$

put $b = -9$ in eq (5)

$$a - 9 = 2$$

$$\boxed{a = 11}$$

put $a = 11$ in eq (4)

$$c = -2 - a$$

$$c = -2 - 11$$

$$\boxed{c = -13}$$

Question: 27-28

Find all values of x for which A is invertible.

27-

$$A = \begin{bmatrix} x-1 & x^2 & x^4 \\ 0 & x+2 & x^3 \\ 0 & 0 & x-4 \end{bmatrix}$$

A is triangular matrix and triangular matrix is only invertible iff its diagonal entries are not 0.

$$\begin{aligned} x-1 &\neq 0, & x+2 &\neq 0, & x-4 &\neq 0 \\ x &\neq 1, & x &\neq -2, & x &\neq 4 \end{aligned}$$

So A is invertible for all values of x except $x = 1, -2, 4$.

28-

$$A = \begin{bmatrix} x - \frac{1}{2} & 0 & 0 \\ x & x - \frac{1}{3} & 0 \\ x^2 & x^3 & x + \frac{1}{4} \end{bmatrix}$$

A is lower triangular matrix.

$$x - \frac{1}{2} \neq 0, \quad x - \frac{1}{3} \neq 0, \quad x + \frac{1}{4} \neq 0$$

$$x \neq \frac{1}{2}, \quad x \neq \frac{1}{3}, \quad x \neq -\frac{1}{4}$$

So A is invertible for all values of x except $x = \frac{1}{2}, \frac{1}{3}, -\frac{1}{4}$.

Question : no : 31-32

Find a Diagonal matrix A that satisfies the given condition

31-

$$A^5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

let $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

$$A^5 = \begin{bmatrix} a^5 & 0 & 0 \\ 0 & b^5 & 0 \\ 0 & 0 & c^5 \end{bmatrix}$$

comparing entries

$$\begin{aligned} a^5 &= 1, \quad \sqrt[5]{b^5} = \sqrt[5]{-1} \\ \sqrt[5]{a^5} &= \sqrt[5]{1}, \quad b = -1 \\ a &= 1, \end{aligned}$$

$$\begin{aligned} \sqrt[5]{c^5} &= \sqrt[5]{-1} \\ c &= -1 \end{aligned}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{Ans}$$

32-

$$A^{-2} = \begin{bmatrix} a & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

let $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

$$A^{-2} = \begin{bmatrix} a^{-2} & 0 & 0 \\ 0 & b^{-2} & 0 \\ 0 & 0 & c^{-2} \end{bmatrix}$$

compare entries

$$\therefore a^{-2} = \frac{1}{a^2}$$

$$\therefore b^{-2} = \frac{-1}{b^2}$$

$$\frac{1}{a^2} = 9$$

$$\frac{1}{b^2} = 4$$

$$1 = 9a^2$$

$$1 = 4b^2$$

$$\frac{1}{9} = a^2$$

$$\Rightarrow b^2 = \frac{1}{4}$$

$$\Rightarrow \sqrt{a^2} = \sqrt{\frac{1}{9}}$$

$$\sqrt{b^2} = \sqrt{\frac{1}{4}}$$

$$a = \frac{1}{3}$$

$$b = \frac{1}{2}$$

$$\therefore c^{-2} = \frac{1}{c^2}$$

$$\frac{1}{c^2} = 1 \rightarrow 1 = c^2$$

$$\Rightarrow c^2 = 1$$

$$\sqrt{c^2} = \sqrt{1}$$

$$c = 1$$

$$A = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Ans.

Question: 3b

Find 3×3 Diagonal matrix A that satisfy

$$A^2 - 3A - 4I = 0$$

Let $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$

$$A^2 - 3A - 4I = 0$$

$$\begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}^2 - 3 \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x^2 & 0 & 0 \\ 0 & y^2 & 0 \\ 0 & 0 & z^2 \end{bmatrix} - 3 \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$x^2 - 3x - 4 = 0$	$y^2 - 3y - 4 = 0$	$z^2 - 3z - 4 = 0$
$x^2 - 4x + x - 4 = 0$	$y^2 - 4y + y - 4 = 0$	$z^2 - 4z + z - 4 = 0$
$x(x-4) + 1(x-4) = 0$	$y(y-4) + 1(y-4) = 0$	$z(z-4) + 1(z-4) = 0$
$(x+1)(x-4) = 0$	$(y-4)(y+1) = 0$	$(z-4)(z+1) = 0$
$x = -1, x = 4$	$y = -1, y = 4$	$z = 4, z = -1$

$(-1, -1, -1), (-1, 4, 4), (-1, -1, 4)$
 $(-1, 4, -1), (4, -1, -1), (4, -1, 4)$
 $(4, 4, -1), (4, 4, 4)$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Question: 37

Let $A = [a_{ij}]$ be an $n \times n$ matrix.
Determine whether A is symmetric

a-

$$a_{ij} = i^2 + j^2$$

Find $a_{ji} = ?$

$$a_{ji} = j^2 + i^2$$

Since addition is commutative

$$j^2 + i^2 = i^2 + j^2$$

So

$$a_{ji} = i^2 + j^2$$

$$a_{ji} = a_{ij}$$

so A is Symmetric.

b-

$$a_{ij} = i^2 - j^2$$

Find a_{ji}

$$a_{ji} = j^2 - i^2$$

$$a_{ij} = i^2 - j^2$$

Since

$$i^2 - j^2 \neq j^2 - i^2$$

$$a_{ij} \neq a_{ji}$$

So Not Symmetric.

c-

$$a_{ij} = 2i + 2j$$

Find $a_{ji} = ?$

$$a_{ji} = 2j + 2i$$

$$a_{ij} = 2i + 2j$$

Since

$$2i + 2j = 2j + 2i$$

So

$$a_{ji} = 2i + 2j$$

$$a_{ji} = a_{ij}$$

So A is Symmetric

d-

$$a_{ij} = 2i^2 + 2j^3$$

Find $a_{ji} = ?$

$$a_{ji} = 2j^2 + 2i^3$$

$$a_{ij} = 2i^2 + 2j^3$$

Since

$$2i^2 + 2j^3 \neq 2j^2 + 2i^3$$

So

$$a_{ji} \neq a_{ij}$$

So not symmetric

Question: 39

Find upper triangular matrix

$$A^3 = \begin{bmatrix} 1 & 30 \\ 0 & -8 \end{bmatrix}$$

Let

$$A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

$$A^3 = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

$$A^3 = \begin{bmatrix} a^2 + 0 & ab + bc \\ 0 + 0 & 0 + c^2 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

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$$A^3 = \begin{bmatrix} a^2 & ab+bc \\ 0 & c^2 \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

$$A^3 = \begin{bmatrix} a^3+0 & a^2b+abc+bc^2 \\ 0+0 & 0+c^3 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} a^3 & a^2b+abc+bc^2 \\ 0 & c^3 \end{bmatrix}$$

Equations:

$a^3 = 1$	$c^3 = -8$
$\sqrt[3]{a^3} = \sqrt[3]{1}$	$\sqrt[3]{c^3} = \sqrt[3]{-8}$
$a = 1$	$c = -2$

$$a^2b + abc + bc^2 = 30$$

$$(1)^2b + (1)b(-2) + b(-2)^2 = 30$$

$$b - 2b + 4b = 30$$

$$3b = 30$$

$$b = 10$$

$$A = \begin{bmatrix} 1 & 10 \\ 0 & -2 \end{bmatrix} \text{ Ans}$$

Question: 41:

Fill entries so matrix A is skew symmetric.

a-

$$A = \begin{bmatrix} x & x & 4 \\ 0 & x & x \\ -x & -1 & x \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 0 & 1 \\ -4 & -1 & 0 \end{bmatrix}$$

b-

$$A = \begin{bmatrix} x & 0 & x \\ x & x & -4 \\ 8 & x & x \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 & -8 \\ 0 & 0 & -4 \\ 8 & 4 & 0 \end{bmatrix}$$

Question: 42

Find all values of a, b, c and d for which A is skew-symmetric.

$$A = \begin{bmatrix} 0 & 2a-3b+c & 3a-5b+5c \\ -2 & 0 & 5a-8b+6c \\ -3 & -5 & d \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2a-3b+c & 3a-5b+5c \\ -2 & 0 & 5a-8b+6c \\ -3 & -5 & d \end{bmatrix} = \begin{bmatrix} 0 & 2 & 3 \\ -(2a-3b+c) & 0 & 5 \\ -(3a-5b+c) & -(5a-8b+6c) & -d \end{bmatrix}$$

$$2a-3b+c=2$$

$$-(3a-5b+c) = -3$$

$$2a-3b+c=2 \rightarrow (1)$$

$$3a-5b+c=3 \rightarrow (2)$$

$$5a-8b+6c=5 \rightarrow (3)$$

$$d = -d \rightarrow (4)$$

$$d = -d$$

$$d + d = 0$$

$$2d = 0$$

$$\boxed{d = 0}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 1 & 2 \\ 3 & -5 & 1 & 3 \\ 5 & -8 & 6 & 5 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -\frac{3}{2} & \frac{1}{2} & 1 \\ 3 & -5 & 1 & 3 \\ 5 & -8 & 6 & 5 \end{array} \right] \begin{array}{l} R_1 \\ \times 2 \\ R_2 \\ - 3R_1 \\ R_3 \\ - 5R_1 \end{array}$$

$$\begin{bmatrix} 1 & -3/2 & 1/2 & 1 \\ 0 & -1/2 & 7/2 & 0 \\ 0 & -1/2 & 7/2 & 0 \end{bmatrix} \begin{matrix} \\ R_3 - 3R_1 \\ R_5 - 5R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & -3/2 & 1/2 & 1 \\ 0 & 1 & -7 & 0 \\ 0 & -1/2 & 7/2 & 0 \end{bmatrix} \begin{matrix} \\ -2R_2 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3/2 & 1/2 & 1 \\ 0 & 1 & -7 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_3 + \frac{1}{2}R_2 \\ \\ \end{matrix}$$

Leading var = a, b Let

Free var $\rightarrow c$ $c = t$

$$a - 3/2 b + 1/2 c = 1$$

$$a - 3/2 b + 1/2 t = 1$$

$$a - 3/2 (7t) + 1/2 t = 1$$

$$a = 1 + 21/2 t - 1/2 t$$

$$a = \frac{2 + 21t - t}{2}$$

$$a = \frac{2 + 20t}{2}$$

$$a = 1 + 10t$$

$$b - 7c = 0$$

$$b = 7t$$

$$c = t$$

$$d = 0$$

Question: 44

Prove that every square matrix A can be expressed as the sum of a symmetric matrix and skew-symmetric matrix.

$$A = \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T)$$

$$\text{Let } P = \frac{1}{2}(A+A^T)$$

$$\text{Let } Q = \frac{1}{2}(A-A^T)$$

P is Symmetric Matrix

$$P^T = \left[\frac{1}{2}(A+A^T) \right]^T$$

$$P^T = \frac{1}{2}(A^T + (A^T)^T)$$

$$P^T = \frac{1}{2}(A^T + A)$$

$$P^T = \frac{1}{2}(A+A^T)$$

$$P^T = P$$

Q is skew Symmetric

$$Q^T = \left[\frac{1}{2}(A-A^T) \right]^T$$

Hence proved

$$Q^T = \frac{1}{2}(A^T - (A^T)^T)$$

$$Q^T = \frac{1}{2}(A^T - A)$$

$$Q^T = - \left[\frac{1}{2}(A-A^T) \right]$$

$$Q^T = -Q$$